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IENG435
HEURISTIC METHODS IN PROBLEM SOLVING
TERM PROJECT

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Contents

1. Method	3
2. Pilot Study	3
2.1 Determination of Parameters	4
3. Main Experiment	4
3.1 Determination of Parameters	4
3. Testing the Parameters of the Main Experiment on Other Datasets	7
3.1 Optimum Solution of Bays.29 Dataset.	7
3.2 Optimum Solution of Att48 Dataset....	8
4. Conclusion.	8

Parameter Optimization and Experimental Design with Heuristic Algorithm for the Traveling Salesman Problem

1. Method

Factorial experimental design was used as the method in the studies conducted. Factorial design of experiments is a systematic method in which multiple parameters are investigated simultaneously. Different values of each parameter are tested, and by testing all combinations, both individual effects and interactions between parameters are analyzed. In this way, the most suitable parameter combinations can be determined more effectively.

2. Pilot Study

The numbers in this table are an output of the full factorial experimental design used within the scope of the pilot study. Each row shows the result of a different parameter combination.

It was conducted to determine which combination of population size and number of iterations yields the best results, and at the same time to analyze the balance between execution time (time complexity) and solution quality.

It is a preliminary study conducted to fix the population and iteration count in the main experimental study.

Table 1

Pop Size	Iterations	Best Fitness	Duration (s)
10	10000	27350.9243	0.000389338
10	50000	26675.0412	0.00019002
10	100000	26427.2761	0.000178099
20	10000	27297.5611	0.000345469
20	50000	26857.2807	0.000354052
20	100000	27436.8897	0.000345945
30	10000	26835.3217	0.000527859
30	50000	25625.6358	0.000514746
30	100000	26260.1952	0.000522137

2.1 Determination of Parameters

Population size: Having more individuals than necessary in the population prolongs the solution time, while a small number of individuals may cause the population not to reach the desired solution value.

Iteration count: In genetic algorithms, the number of iterations determines how many generations the algorithm will evolve. This parameter is a critical factor directly affecting the success and efficiency of the algorithm. A high number of iterations converges to reaching the optimum solution (does not guarantee it), but this requires excessive processing time and, by the nature of the Genetic Algorithm, it may get stuck in local optima.

Parameters:

Parameter	Description	Experimental Values
Pop Size	Population size in the genetic algorithm	10, 20, 30
Iterations	Number of iterations (generations) the algorithm runs	10.000, 50.000, 100.000
Best Fitness	Total distance of the best solution route found	(Varies depending on the solution)
Duration (s)	Computation time (in seconds)	(Measured automatically)

3. Main Experiment

In this study, a factorial experimental design was adopted to optimize the parameters affecting the performance of the genetic algorithm on the Berlin52 problem. Experimental designs were carried out to evaluate the effects of various parameter combinations on solution quality. Experimental designs were made according to the top 2 best fitness values of the pilot study.

Elitism rate was determined as 0.1, 0.2, 0.3 parameters.

Crossover rate was determined as 0.5, 0.7, 0.9 parameters.

Mutation rate was determined as 0.01, 0.05, 0.1 parameters.

3.1 Determination of Parameters

Elitism rate: Refers to the direct transfer of a certain portion of the best individuals found in each generation to the next generation. In this way, high-quality solutions are prevented from being lost due to genetic operators involving randomness (crossover and mutation). The elitism rate determines how many individuals this transfer will cover and is generally defined as a certain percentage of the population. If the elitism rate is kept too high, diversity in the population may decrease. This limits the algorithm's capacity to explore new and different solutions and increases the probability of the algorithm getting stuck in local optima. Therefore, the elitism rate must be balanced.

Crossover rate: Genetic algorithms (GA) are a heuristic optimization method that mimics biological evolution processes in order to obtain better individuals in the solution space. In these algorithms, crossover is one of the fundamental operators used to produce new individuals from existing solutions. The crossover rate determines how many individuals in the population will undergo recombination through this process and directly affects the exploration capability of the algorithm.

A high crossover rate can contribute to avoiding local optima by enabling a wider search in the solution space. However, when this rate is too high, it may become difficult to preserve elite individuals and the convergence speed of the algorithm may decrease. On the other hand, a low crossover rate provides a more conservative search, but by reducing diversity, it can increase the risk of premature convergence. Therefore, the crossover rate should be chosen carefully; a balanced value should be determined depending on the problem structure and other parameters.

Mutation rate: Mutation is one of the fundamental operators in genetic algorithms that introduces diversity into the population by randomly altering the genetic structure of individuals. The mutation rate determines the probability of genes in each individual undergoing mutation and plays an important role in both the exploration capacity of the algorithm and reducing the risk of getting trapped in local minima.

A low mutation rate allows the algorithm to converge steadily while preserving existing good solutions. However, in this case, the solution space may not be sufficiently explored and the algorithm may get stuck in local optima. A high mutation rate increases population diversity, allowing for a broader search, but this increases the randomness of the algorithm, which may lead to the disruption of good solutions and delayed convergence. Therefore, the mutation rate is critical in terms of solution quality and algorithm efficiency, and should be carefully determined in accordance with the problem type.

Table 2

Dataset	PS	Iterations	ER	CR	MR	Seed	Best Fitness
[[1.000e+0	30	100000	0.1	0.5	0.01	0	8680.235325
[[1.000e+0	30	100000	0.1	0.5	0.01	1	8720.248132
[[1.000e+0	30	100000	0.1	0.5	0.01	2	8947.292515
[[1.000e+0	30	100000	0.1	0.5	0.01	3	8997.266686
[[1.000e+0	30	100000	0.1	0.5	0.01	4	9040.712302
[[1.000e+0	30	100000	0.1	0.5	0.01	5	9155.19087
[[1.000e+0	30	100000	0.1	0.5	0.01	6	9179.957901
[[1.000e+0	30	100000	0.1	0.5	0.01	7	9193.509652
[[1.000e+0	30	100000	0.1	0.5	0.01	8	9206.556261
[[1.000e+0	30	100000	0.1	0.5	0.01	9	9207.838877

In Table 2, as a result of the pilot study, the population size was fixed at 30 and the number of iterations at 1e5. Crossover rate, elitism rate, and mutation rate were determined in a balanced manner for the problem to form the parameters. 10 seeds were randomly determined and included in the code, and run at each iteration. The best fitness value was found to be 8680.

Table 3

Dataset	PS	Iterations	ER	CR	MR	Seed	Best Fitness
[[1.000e+0	30	50000	0.1	0.5	0.01	0	9636.761626
[[1.000e+0	30	50000	0.1	0.5	0.01	1	9946.128147
[[1.000e+0	30	50000	0.1	0.5	0.01	2	10206.75217
[[1.000e+0	30	50000	0.1	0.5	0.01	3	10265.24735
[[1.000e+0	30	50000	0.1	0.5	0.01	4	10300.1316
[[1.000e+0	30	50000	0.1	0.5	0.01	5	10341.22756
[[1.000e+0	30	50000	0.1	0.5	0.01	6	10379.7188
[[1.000e+0	30	50000	0.1	0.5	0.01	7	10406.72139
[[1.000e+0	30	50000	0.1	0.5	0.01	8	10447.71086
[[1.000e+0	30	50000	0.1	0.5	0.01	9	10543.49357

In Table 3, as a result of the pilot study, the population size was fixed at 30 and the number of iterations at 5e4. Crossover rate, elitism rate, and mutation rate were determined in a balanced manner for the problem to form the parameters. 10 seeds were randomly determined and included in the code, and run at each iteration. The best fitness value was found to be 9636.

Comparison of Table 2 and Table 3

When the population size was 30 and the iteration number was 1e5, it gave better results together with the other parameters. That is, as the iteration increased, the rate of approaching the optimum solution increased.

In Table 4, keeping the iteration number fixed at $1e5$, a population size of 40 was also tested instead of 30, but it moved away from the optimum solution.

Table 4

Dataset	PS	Iterations	ER	CR	MR	Seed	Best Fitness
[[1.000e+0	40	100000	0.1	0.5	0.01	0	9040.84355
[[1.000e+0	40	100000	0.1	0.5	0.01	1	9072.227874
[[1.000e+0	40	100000	0.1	0.5	0.01	2	9216.29996
[[1.000e+0	40	100000	0.1	0.5	0.01	3	9264.552143
[[1.000e+0	40	100000	0.1	0.5	0.01	4	9270.091201
[[1.000e+0	40	100000	0.1	0.5	0.01	5	9270.882899
[[1.000e+0	40	100000	0.1	0.5	0.01	6	9319.111397
[[1.000e+0	40	100000	0.1	0.5	0.01	7	9341.108801
[[1.000e+0	40	100000	0.1	0.5	0.01	8	9374.116915
[[1.000e+0	40	100000	0.1	0.5	0.01	9	9384.094421

3. Testing the Parameters of the Main Experiment on Other Datasets

3.1 Optimum Solution of Bays.29 Dataset

Table 5

PS	Iterations	ER	CR	MR	Seed	Best Fitness
30	100000	0.1	0.5	0.01	0	9074.148048
30	100000	0.1	0.5	0.01	1	9074.148048
30	100000	0.1	0.5	0.01	2	9074.148048
30	100000	0.1	0.5	0.01	3	9112.542236
30	100000	0.1	0.5	0.01	4	9112.542236
30	100000	0.1	0.5	0.01	5	9120.338841
30	100000	0.1	0.5	0.01	6	9170.054128
30	100000	0.1	0.5	0.01	7	9170.303781
30	100000	0.1	0.5	0.01	8	9170.303781
30	100000	0.1	0.5	0.01	9	9181.744782

3.2 Optimum Solution of Att48 Dataset

Table 6

Dataset	PS	Iterations	ER	CR	MR	Seed	Best Fitness
[[1.000e+0	30	100000	0.1	0.5	0.01	0	36701.84162
[[1.000e+0	30	100000	0.1	0.5	0.01	1	36745.97998
[[1.000e+0	30	100000	0.1	0.5	0.01	2	37595.82539
[[1.000e+0	30	100000	0.1	0.5	0.01	3	37610.21573
[[1.000e+0	30	100000	0.1	0.5	0.01	4	38029.64327
[[1.000e+0	30	100000	0.1	0.5	0.01	5	38268.71783
[[1.000e+0	30	100000	0.1	0.5	0.01	6	38583.22852
[[1.000e+0	30	100000	0.1	0.5	0.01	7	39036.02173
[[1.000e+0	30	100000	0.1	0.5	0.01	8	39055.80561
[[1.000e+0	30	100000	0.1	0.5	0.01	9	39088.01976

4. Conclusion

When the population size is fixed at 30 and the number of iterations at $1e5$ for all datasets; it has generally been observed that the best results are obtained when the elitism rate is determined as 0.1, the crossover rate as 0.5, and the mutation rate as 0.01.